Astrea Academy Trust
INSPIRING BEYOND MEASURE

Name	
Class	
Tutor Group	

YEAR 11 MATHEMATICS

Unit 1: Further Algebra (PART 1)

PART 1: Rearranging Formulae, Simultaneous Equations (including Quadratic Simultaneous Equations)

$$\frac{x+5}{2x+10}$$

$$x_{n+1} = \sqrt{x_n + 2}$$

$$f(x) = 2x + 3$$

$$f^{-1}(x) = \frac{x-3}{2}$$

KEY VOCABULARY LIST	
Fraction	Fractions represent equal parts of a whole or a collection.
Numerator	The top number in a fraction, represents the parts.
Denominator	The bottom number in a fraction, represents the whole.
Function	A special relationship where each input has a single output, usually written as $f(x)$.
Inverse Function	An inverse function is a function that undoes the action of the original function.
Composite Function	Composite functions are where more than one function is applied to an input, in a specific order.
Formula	A formula is special equation that links two or more variables.
Iteration	Iteration means repeatedly carrying out a process. In this unit, start with an initial value and substitute this into the iteration formula to obtain a new value, then use the new value for the next substitution, and so on.

KEY VOCABULARY LIST	
Solve	To solve means to find a solution. Simultaneous equations have **two** solutions.
Quadratic simultaneous equations	Quadratic simultaneous equations are two or more equations that share variables which include x^2 and y^2
Constant	In algebra, a constant is a number on its own, or sometimes a letter such as a, b or c that stands for a fixed number.
Variable	Variable, in algebra, a symbol (usually a letter) standing in for an unknown numerical value in an equation.
Linear Function	- those whose graph is a straight line. - has the following form. y = mx + c. - has one independent variable and one dependent variable. The independent variable is x and the dependent variable is y.
Simultaneous Equations	Two or more equations that share variables. Example: two equations that share the variables x and y: x + y = 6 −3x + y = 2
Elimination Method	Eliminate" means to remove: this method works by removing variables until there is just one left. The idea is that we **can safely**: **multiply** an equation by a constant (except zero), **add** (or subtract) an equation on to another equation
Substitution Method	These are the steps: Write one of the equations so it is in the style "variable = ..." Replace (i.e. substitute) that variable in the other equation(s). Solve the other equation(s)

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BOOKLET STRUCTURE

- **Start of subtopic:**
  - Read and annotate the information in the Frayer model or introduction text
  - I do (teacher led example).
  - We do (teacher and scholar led example).
  - You do (independent scholar example).
  - Whiteboard check.
  - Independent practice and mixed practice.

- **End of subtopic:**
  - GCSE questions – to ensure you are confident with the style you will encounter in your exams

Subtopic 1 – Rearranging Formulae/ Changing the Subject

LEARNING OUTCOME: To be fluent when rearranging formulae/ changing the subject

Objectives:

1. To be able to rearrange formulae with one-step
2. To be able to rearrange formulae with two-steps
3. To be able to rearrange formulae requiring many-steps

Keywords:
- Equation
- Subject
- Inverse
- Formulae

Introduction/Recap

1. Rearranging formulae is a lot like solving equations—you still use inverse operations on both sides to
2. keep things balanced. The difference is, instead of finding a number, you're working to make a
3. specific variable (called the *subject*) stand alone. Making a variable the subject means rewriting the
4. equation so that it's isolated on one side, usually the left. This lets you express that variable in terms
5. of the others. This is sometimes called **changing the subject.**

Prior Knowledge Quiz – One Step Rearranging

There are many types of 'rearranging' and 'changing the subject' questions – they can be simple or get very challenging. The Quiz on the pages below should be attempted individually, and your teacher will assess how much recapping is needed before you move on to more advanced algebra topics. The later topics generally rely on you being able to rearrange formulae fluently.

Question 1. Make x the subject of the following:

a) $x + 2p = k$	b) $12 = x - 2y$	c) $-2y = x - 12$	d) $\frac{a-b}{2} = x + c$

Question 2. Make x the subject of the following:

a) $2x = b$	b) $\frac{x}{2} = b$	c) $\frac{x}{c} = b$	d) $xc = b$
e) $ab = cx$	f) $ab = \frac{x}{d}$	g) $y = \frac{x}{bc}$	h) $mx = w + q$

Question 3. Make x the subject of the following:

a) $x^2 = y$	b) $y = x^2$	c) $\sqrt{x} = a$	d) $x^2 = a + 5$
e) $x^2 = 2y$	f) $\sqrt{x} = y + 1$	g) $\sqrt{x} = y^2 + 1$	h) $x^2 = y^2 + 1$

Two step rearranging (addition/ subtraction with division)

	Example 1 – I do…	***Example 2 - We do…***	***Example 3 - You do…***
Steps	Make x the subject of the following:	Make x the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Use inverse operation**s** to isolate the variable. Be careful about the order you do things in.	$y = mx + c$	$y = abx - c$	$y = ax - b$

Independent Practice

Question 1. Make x the subject of the following:

a) $y = ax + b$	b) $y = bx - a$	c) $4x + c = w$	d) $7x + b = 9$

e) $a + 6x = q$	f) $5x - b = c$	g) $a = 2x - c$	h) $dx - t = 8$
i) $s = tx - d$	j) $2x + 2y = p$	k) $y = ax - b$	l) $y = ax - bc$
m) $y = ax - bcd$	n) $y = ax + bcd$	o) $y = d^2ef + ax$	p) $y = -\sqrt{b} + ax$

Two step rearranging (addition/ subtraction with multiplication)

	Example 4 – I do…	***Example 5 - We do…***	***Example 6 - You do…***
Steps	Make y the subject of the following:	Make y the subject of the following:	Make d the subject of the following:
1) Circle the variable you want to make the subject. 2) Use inverse operation**s** to isolate the variable.	$\frac{y - a}{5} = m$	$\frac{y}{5} - a = m$	$\frac{d + a}{4} = s$

Independent Practice

Question 2. Make x the subject of the following:

a) $\frac{x+4}{3} = m$	b) $\frac{x+3}{5} = h$	c) $t = \frac{x-v}{4}$	d) $\frac{x+b}{a} = r$

e) $\frac{x+3}{c} = h$	f) $\frac{w+x}{u} = 3z$	g) $\frac{x+t}{m} = 2c$	h) $\frac{x+q}{p} = p$
i) $h = \frac{x}{2} - f$	j) $\frac{x}{6} - 5 = w$	k) $t = \frac{x}{4} + 1$	l) $y = \frac{x}{3} - 7$

Multi-step rearranging

	Example 7 – I do…	***Example 8 - We do…***	***Example 9 - You do…***
Steps	Make x the subject of the following:	Make y the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Use inverse operation**s** to isolate the variable.	$\frac{7x+b}{3} = 9$	$2y^2 + b = m$	$\frac{\sqrt{x}}{3} + 9 = \mathrm{r}$

Independent Practice

Question 3. Make x the subject of the following:

a) $\frac{7x+m}{11}=1$	b) $\frac{7x^2+m}{11}=1$	c) $5x^2-ab=c$	d) $5x^2+ab=c$
e) $7x+c-b=9$	f) $\frac{p-4+2x}{m}=m$	g) $\frac{x^2}{3}-g=a$	h) $\frac{\sqrt{x}}{a}-d=f$
i) $\frac{\sqrt[3]{x}}{a}-d=f$	j) $\frac{\sqrt{x}}{g}+b=m$	k) $\frac{x^3}{2}=g+5$	l) $\frac{x^2}{10}+b=a^2$
m) $\frac{x^2}{5}+ab=a^2$	n) $\frac{x^3}{5}+ab=a^2$	o) $(m-3)x^2=g$	p) $(m-n)x^2=g$

Rearranging More Complex Formulae

LEARNING OUTCOME: To be able to rearrange more complex formulae

Objectives:

1. To be able to rearrange equations with negative unknowns
2. To be able to rearrange equations with the subject in the denominator
3. To be able to rearrange equations where the subject appears more than once

Keywords:
- Equation
- Subject
- Inverse
- Formulae

Introduction

1. When rearranging more complex equations, you'll use the same basic steps, but things can get a bit
2. trickier. You might have to deal with negative variables, variables in the denominator, or even the
3. subject appearing more than once. In these cases, you may need to multiply both sides to eliminate
4. fractions, move terms carefully, and sometimes factorise expressions to get the subject on its own. It
5. takes a few more steps, but the key is still to use inverse operations and keep both sides balanced.

Changing the subject

Definition

Changing the subject of a formula means rearranging the formula to make a different variable the subject (i.e., on its own on one side of the equals sign).

Characteristics

- Involves inverse operations (e.g., undoing addition with subtraction) and works step-by-step like solving an equation, keeping the equation balanced on both sides.
- The subject is the variable you're isolating.

Examples

$A = lw$ has been rearranged to $l = \frac{A}{w}$ to make l the subject.

$v = u + at$ has been rearranged to $t = \frac{v-u}{a}$ to make t the subject.

Non - Examples

"Solve $3x + 2 = 11$"

This is solving, not changing the subject of a formula.

Rearranging with negative unknowns

	Example 1 – I do…	*Example 2 - We do…*	*Example 3 - You do…*
Steps	Make y the subject of the following:	Make x the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Use inverse operations to isolate the variable. 3) Remember to divide by the coefficient of the subject (this will mean you are dividing by a negative).	$x = a - 3y$	$u = \frac{-x}{v + w}$	$5b - x = a$ $x = -\frac{y}{a} - 4$

Independent Practice

Question 1. Make x the subject of the following:

a) $a - x = y$	b) $b - x = c$	c) $\mathrm{y} = -\frac{x}{a}$
d) $-\frac{x}{m} = \mathrm{k}$	e) $m = -\frac{x}{a^2}$	f) $-10ax = b$
g) $-\frac{x}{5} - a = m$	h) $-\frac{x}{3} + a = b$	i) $f - \frac{x}{c} = q$
j) $\frac{-a-x}{5} = m$	k) $-5x + a = m$	l) $-px - e = 5t$
m) $t = -\frac{2x}{3} + b$	n) $-ax + y = 2r$	o) $-7x^3 - b = 9$

Rearranging with the subject in the denominator

	Example 4 – I do…	*Example 5 - We do…*	*Example 6 - You do…*
Steps	Make y the subject of the following:	Make y the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Multiply both sides by the denominator (subject). 3) Use inverse operations to isolate the variable.	$\frac{p}{y} = x$	$\frac{m - w}{y} = a^2$	$r^2 = \frac{p + q}{x}$
			$\frac{a}{x} = b + 2$

Independent Practice

Question 2. Make w the subject of the following:

a) $\frac{g}{w} = m$	b) $a^2 = \frac{p}{w}$	c) $b + c = \frac{a}{w}$
d) $\frac{x + 2}{w} = k - 4$	e) $x + y = \frac{a + b}{w}$	f) $\frac{c - x}{w} = m - n$

Rearranging where the subject appears twice

Introduction

Sometimes, when rearranging a formula, the variable you want to make the subject appears more than once. In this case, you can't isolate it right away—you first need to collect all the terms containing that variable on one side of the formula. Then, you factorise by taking the variable out, which means writing it once, multiplied by a bracket. This makes it possible to divide and fully isolate the subject. It's an extra step, but it follows the same logic: work step by step to get the subject by itself. In this section you will practice a series of increasingly difficult question types that involve factorising, while incorporating all of the previous rearranging knowledge you have acquired.

	Example 7 – I do…	***Example 8 - We do…***	***Example 9 - You do…***
Steps	Make x the subject of the following:	Make x the subject of the following:	Make x the subject of the following:
1) Circle the subject that we want to isolate. 2) Factorise the subject. 3) Divide both sides of the equation.	$ax + bx = y$	$ax - bx = y$	$bx - ax = y$

Independent Practice

Question 3. Make b the subject of the following:

a) $bh + ab = 9$	b) $5b + ab = 9$	c) $bh - ab = 9$	d) $6b - ab = 9$

	Example 10 – I do…	*Example 11 - We do…*	*Example 12 - You do…*
Steps	Make a the subject of the following:	Make a the subject of the following:	Make a the subject of the following:
1) Circle the subject that we want to isolate. 2) Rearrange to get the subject on **one side.** 3) Factorise the subject. 4) Divide both sides of the equation.	$ax + 2y = 5 + am$	$ab + 3y = 7y + ak$	$ax - 3y = 9a - x$

Independent Practice

Question 4. Make a the subject of the following:

a) $ax + 3y = 5 + a$	b) $ax + 2y = 5 - am$	c) $5y + 2a = x + am$	d) $-2a + 4 = x + am$

	Example 13 – I do…	*Example 14 - We do…*	*Example 15 - You do…*
Steps	Make a the subject of the following:	Make a the subject of the following:	Make a the subject of the following:
1) Circle the subject that we want to isolate. 2) Expand and simplify the brackets. 3) Rearrange to get the subject on **one side.** 4) Factorise the subject. 5) Divide both sides of the equation.	$m(a+y)=x(a+b)$	$3(a+x)=x(a+b)$	$m(a+y)=x(b-a)$

Independent Practice

Question 6. Make a the subject of the following:

a) $7(a+y)=x(a+b)$	b) $5(a-y)=x(a+b)$	c) $6(x-a)=x(a+b)$

	Example 16 – I do…	*Example 17 - We do…*	*Example 18 - You do…*
Steps	Make x the subject of the following:	Make x the subject of the following:	Make x the subject of the following:
1) Circle the subject that we want to isolate. 2) Multiply both sides by the denominator. 3) Expand and simplify the brackets. 4) Rearrange to get the subject on **one side.** 5) Factorise the subject. 6) Divide both sides of the equation.	$y = \frac{x+2}{x-2}$	$y = \frac{x+2}{x-3}$	$y = \frac{x-2}{x+2}$

Independent Practice

Question 7. Make x the subject of the following:

a) $y = \frac{x+3}{x-2}$	b) $y = \frac{x-3}{x+2}$	c) $y = \frac{x+5}{x-2}$

Q1.

Make a the subject of the formula $p = 3a - 9$

.....

(Total for question = 2 marks)

Q2.

$v^2 = u^2 + 2as$

$u = 12 \quad a = -3 \quad s = 18$

(a) Work out a value of v.

.....

(2)

(b) Make s the subject of $v^2 = u^2 + 2as$

.....

(2)

(Total for question = 4 marks)

Q3.

$T = 4m^2 - 11$

(a) Work out the value of T when $m = -3$

$T =$ .....

(2)

(b) Make p the subject of the formula $d = 3p + 4$

.....

(2)

(Total for question = 4 marks)

Q4.

Make k the subject of the formula $y = \sqrt{2m - k}$

.....

(Total for question = 2 marks)

Q5.

Make m the subject of:

$$k = p + \frac{2m}{5}$$

.....

(Total for question = 3 marks)

Q6.

$$T = \frac{q}{2} + 5$$

Here is Spencer's method to make q the subject of the formula.

$$2 \times T = q + 5$$

$$q = 2T - 5$$

What mistake did Spencer make in the first line of his method?

.....

.....

.....

(Total for question = 1 mark)

Q7.

Make m the subject of the formula

$$f = \frac{3m + 4}{m - 1}$$

.....

(Total for question = 3 marks)

Q8.

Make f the subject of the formula

$$d = \frac{3(1 - f)}{f - 4}$$

.....

(Total for question = 4 marks)

Q9.

Make p the subject of the formula

$$t = \frac{2(2p-3)}{5-2p}$$

.....

(Total for question = 4 marks)

Q10.

Make x the subject of the formula

$$y = \frac{4(2x-7)}{5x+3}$$

.....

(Total for question = 4 marks)

Subtopic 2 – Recap: Solving Quadratic Equations

LEARNING OUTCOME: Recap the core skill of solving quadratics.

Objectives:

1. To recap the three main methods of solving quadratics:
   - Factorise
   - Use the quadratic formula
   - Complete the square

Recap: What is a Quadratic and why are they so important in GCSE maths?

A quadratic expression has the form,

$$ax^2 + bx + c,$$

where a, b and c are constants.

Any expression is called quadratic if there is only one algebraic variable in the expression, the highest power of that variable is 2 and there are no variables to powers below 0 (i.e. x^{-1}).

Quadratic equations are where expressions are set to a value, and to solve them means to find the value(s) that make the equation true. Quadratic equations are special in that they may have 2 solutions (or sometimes 1, or sometimes even none).

For the 'further algebra' topics at Higher tier GCSE – such as 'Algebraic fractions', 'Functions' and 'Simultaneous equations', you will often construct a quadratic equation that eventually needs solving. For that reason, it is crucial that you are fluent in the main methods of solving equations, especially factorising and using the quadratic formula. (You may also have been shown how to solve by completing the square).

PRIOR KNOWLEDGE QUIZ

1. Solve each equation **by factorising first**.
You will have to check the equation is in the form $x^2 + bx + c = 0$

a. $x^2 + 4x + 4 = 0$	b. $x^2 + 5x + 6 = 0$	c. $x^2 + 10x + 24 = 0$
d. $x^2 + 20 = 9x$	e. $x^2 = 12x - 20$	f. $15 = 8x - x^2$

PRIOR KNOWLEDGE QUIZ (CONT.)

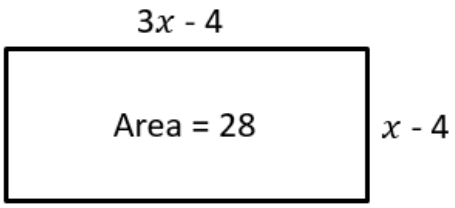
2. Solve each equation, **by factorising first**.
You will have to check the equation is in the form $ax^2 + bx + c = 0$

a) $2x^2 + 3x - 35 = 0$	b) $4x^2 - 3x - 22 = 0$	c) $3x^2 + 4x - 4 = 0$
d) $2x^2 = 3x + 5$	e) $2x^2 + 1 = 3x$	f) $3x^2 + 6x = x - 2$

3. Solve each equation **using the quadratic formula**.
You will have to check the equation is in the form $ax^2 + bx + c = 0$

a) $x^2 + 6x + 3 = 0$	b) $2x^2 + 7x + 1 = 0$	c) $x^2 - 7x + 1 = 0$

a) Determine the value of x:



The diagram shows a trapezium.

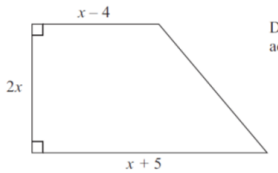


Diagram **NOT** accurately drawn

All the measurements are in centimetres.

The area of the trapezium is 351 cm^2.

(*a*) Show that $2x^2 + x - 351 = 0$

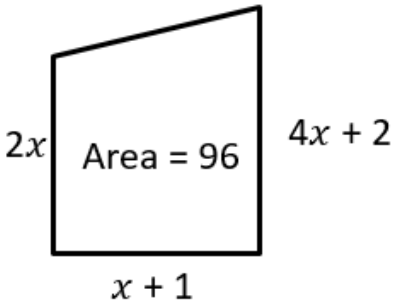
(2)

(*b*) Work out the value of x.

.....

(3)

c) Determine the value of x



Solve $\frac{5(2x+1)^2}{4x+5} = 5x - 1$

Question 1: Solve these **increasingly difficult** quadratic equations:

a) $x = -3, x = -5$	b) $x = -4, x = -1$	c) $x = -5, x = 2$
d) $x = 10, x = -3$	e) $x = 7, x = 1$	f) $x = 8, x = -8$
g) $x = -0.5, x = -3$	h) $x = -5, x = 1/3$	i) $x = -1, x = -3$
j) $x = -2, x = 5$	k) $x = -2, x = -3$	l) $x = 2, x = 3$

Question 2: Solve each of the equations below

a)	$x = -2$ and $x = -\frac{1}{2}$	b)	$x = -2\frac{1}{2}$ and $x = -1$	c)	$x = -\frac{2}{5}$ and $x = -1$
d)	$x = -4$ and $x = -4\frac{1}{2}$	e)	$x = -\frac{3}{5}$ and $x = -4$	f)	$x = -2$ and $x = -\frac{1}{3}$
g)	$x = -\frac{1}{3}$ and $x = -1$	h)	$x = \frac{1}{2}$ and $x = -4$	i)	$x = 2$ and $x = -1\frac{1}{2}$
j)	$x = -3$ and $x = -\frac{2}{7}$	k)	$x = -\frac{2}{3}$ and $x = 1$	l)	$x = \frac{1}{5}$ and $x = 3$
m)	$x = 1$ and $x = -\frac{4}{3}$	n)	$x = 1\frac{1}{2}$ and $x = 5$	o)	$x = 2$ and $x = 1\frac{1}{7}$
p)	$x = -9\frac{1}{2}$ and $x = 2$	q)	$x = 1\frac{1}{5}$ and $x = 5$	r)	$x = 6$ and $x = -2\frac{2}{3}$

(c) $5x^2 + 7x + 2 = 0$

(f) $3x^2 + 7x + 2 = 0$

(i) $2x^2 - x - 6 = 0$

(l) $5x^2 - 16x + 3 = 0$

(o) $7x^2 - 22x + 16 = 0$

(r) $3x^2 - 10x - 48 = 0$

Question 3: Solve each of the equations below

a)	$x = -1\frac{1}{2}$ and $x = -\frac{1}{2}$	b)	$x = -3\frac{1}{2}$ and $x = \frac{1}{2}$	c)	$x = \frac{3}{4}$ and $x = 2$
d)	$x = -5$ and $x = -\frac{1}{6}$	e)	$x = 4\frac{1}{2}$ and $x = -\frac{1}{2}$	f)	$x = 1\frac{1}{2}$ and $x = -\frac{1}{4}$
g)	$x = \frac{1}{10}$ and $x = 1$	h)	$x = -4\frac{1}{2}$ and $x = -\frac{2}{3}$	i)	$x = -\frac{2}{3}$ and $x = 1\frac{1}{3}$
j)	$x = -2\frac{1}{2}$ and $x = 3\frac{1}{2}$	k)	$x = -\frac{3}{4}$ and $x = -\frac{4}{3}$	l)	$x = \frac{5}{14}$ and $x = -2$
m)	$x = -2\frac{1}{2}$ and $x = \frac{1}{3}$	n)	$x = -\frac{1}{2}$ and $x = 2\frac{1}{3}$	o)	$x = \frac{3}{8}$ and $x = 1\frac{1}{2}$

(c) $4x^2 - 11x + 6 = 0$

(f) $8x^2 - 10x - 3 = 0$

(i) $9x^2 - 6x - 8 = 0$

(l) $14x^2 + 23x - 10 = 0$

(o) $16x^2 - 30x + 9 = 0$

a)	$x = \pm 1\frac{1}{2}$	b)	$x = \pm 5\frac{1}{2}$	c)	$x = \pm 1\frac{1}{4}$	d)	$x = \pm \frac{1}{6}$
e)	$x = \pm 4\frac{2}{3}$	f)	$x = \pm \frac{7}{10}$	g)	$x = \pm 15$	h)	$x = \pm 1\frac{5}{8}$

LEARNING OUTCOME: Solve simultaneous equations in two variables by the graphical method

Objectives:

1. Solve graphical simultaneous equations from a given graph.
2. Solve graphical simultaneous equations by plotting one or both lines.

Keywords:
- Simultaneous equations
- Linear equations
- Equivalent equations
- Intersection

Graphical simultaneous equations

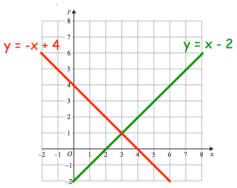
Definition

Two equations that form straight lines that **intersect at one point.**

Characteristics

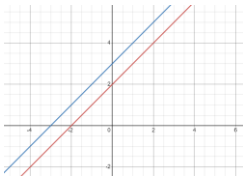
Linear simultaneous equations are two straight line equations that have an intersection point (this means they cross on the graph).

Examples



These are simultaneous equations because they **intersect** at a point.

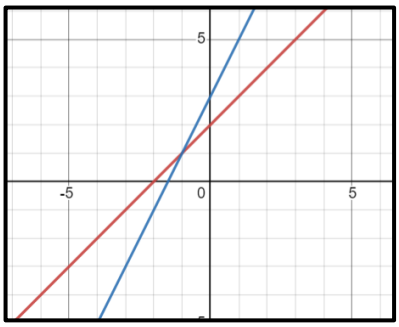
Non - Examples



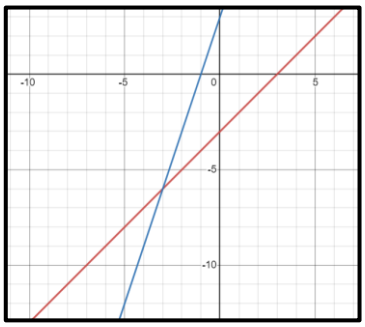
These equations do not intersect, therefore they are not simultaneous.

KNOWLEDGE CHECK

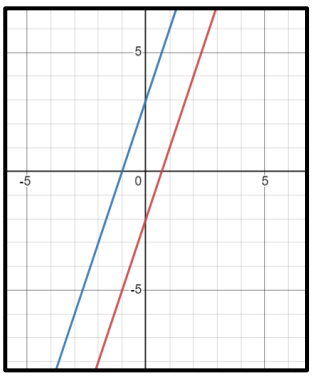
Do these graphs intersect?



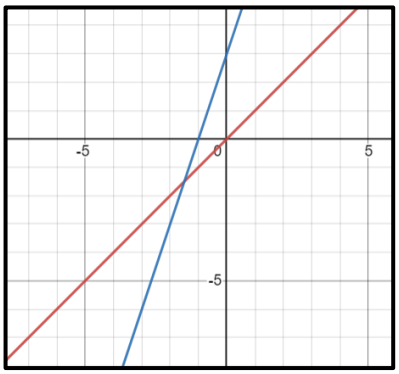
Y ☐ N ☐



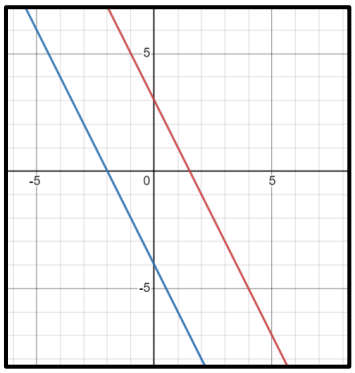
Y ☐ N ☐



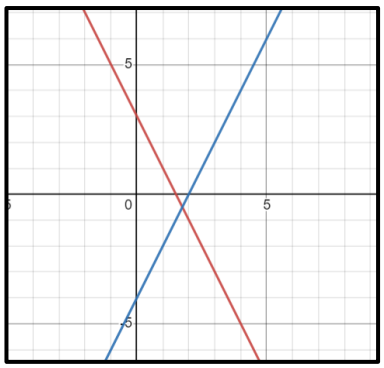
Y ☐ N ☐



Y ☐ N ☐



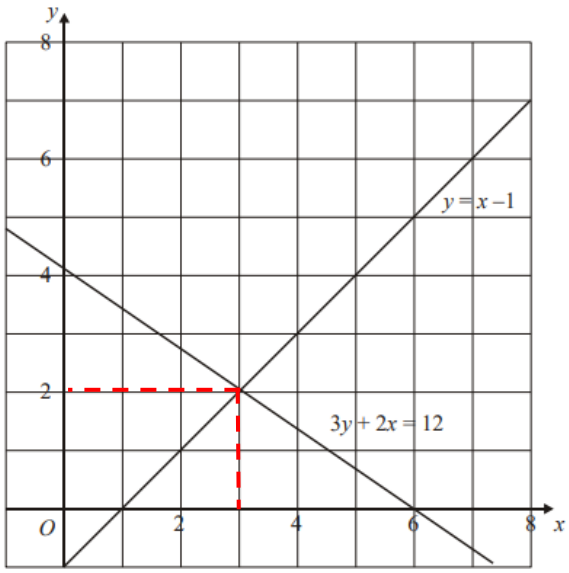
Y ☐ N ☐



Y ☐ N ☐

Example 1 - I do…

The graphs of the straight lines with equations $3y + 2x = 12$ and $y = x - 1$ have been drawn on the grid.



Use the graph to solve the simultaneous equations
$3y + 2x = 12$ and $y = x - 1$

Step 1:

Find the **intersection** point.

Step 2:

Draw a line from the intersection point to the x –axis.

Step 3:

Draw a line from the intersection point to the y-axis.

Step 4:

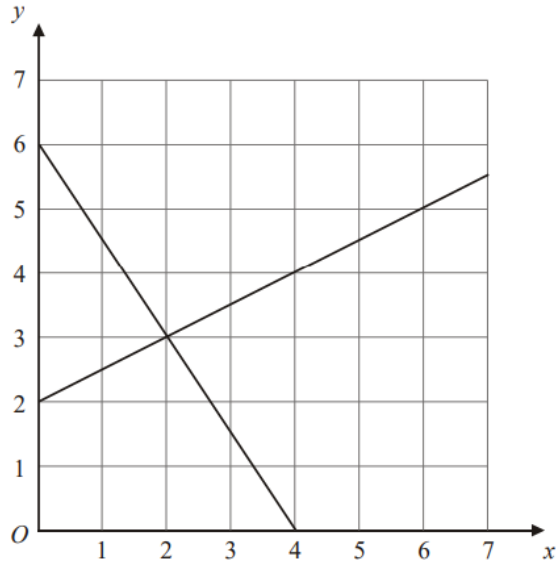
Write the x and y coordinates as the solution.

$x =$

$y =$

Example 2 – We do…

The graphs of the straight lines with equations $y + \frac{1}{2}x + 2$ and $2y + 3x = 12$ have been drawn on the grid.



Use the graph to solve the simultaneous equations
$y + \frac{1}{2}x + 2$ and $2y + 3x = 12$

Step 1:

Find the **intersection** point.

Step 2:

Draw a line from the intersection point to the x –axis.

Step 3:

Draw a line from the intersection point to the y-axis.

Step 4:

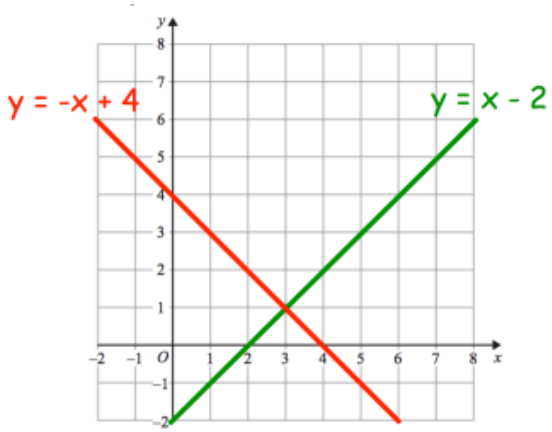
Write the x and y coordinates as the solution.

$x =$

$y =$

Example 3 - You do…

The graphs of the straight lines with equations $y = -x + 4$ and $y = x - 2$ have been drawn on the grid.



Use the graph to solve the simultaneous equations
$y = -x + 4$ and $y = x - 2$

Step 1:

Find the **intersection** point.

Step 2:

Draw a line from the intersection point to the x –axis.

Step 3:

Draw a line from the intersection point to the y-axis.

Step 4:

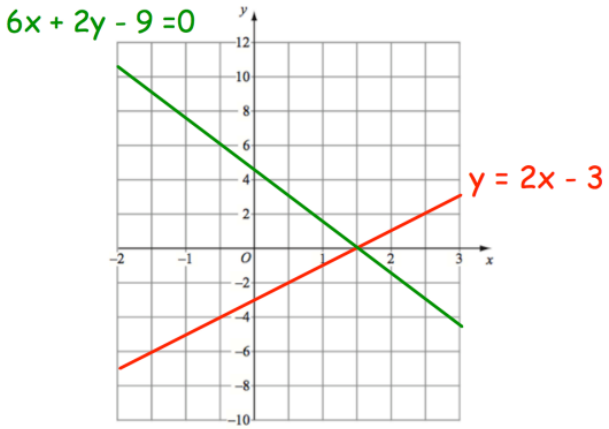
Write the x and y coordinates as the solution.

$x =$

$y =$

Example 4 – You do…

The graphs of the straight lines with equations $6x + 2y - 9 = 0$ and $y = 2x - 3$ have been drawn on the grid.



Use the graph to solve the simultaneous equations
$6x + 2y - 9 = 0$ and $y = 2x - 3$

Step 1:

Find the **intersection** point.

Step 2:

Draw a line from the intersection point to the x –axis.

Step 3:

Draw a line from the intersection point to the y-axis.

Step 4:

Write the x and y coordinates as the solution.

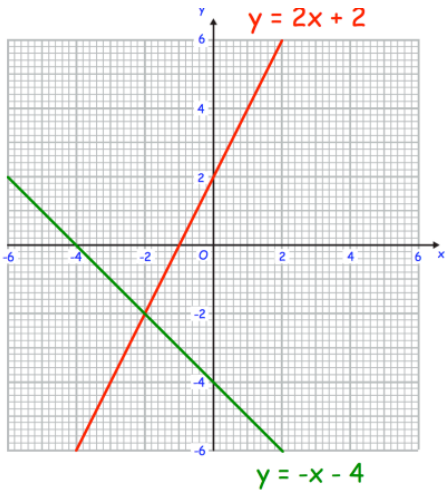
$x =$

$y =$

INDEPENDENT PRACTICE

1) Solve the simultaneous equations graphically

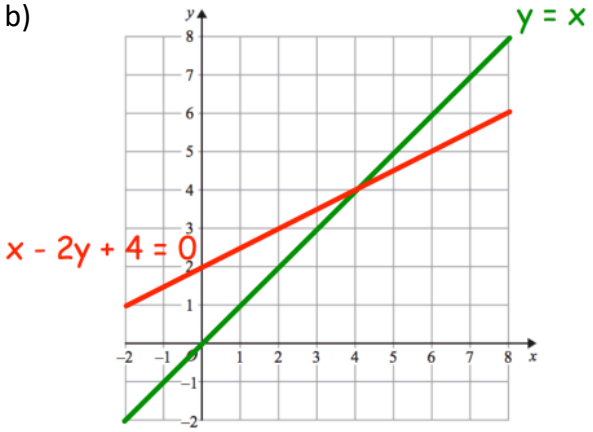
a)



$x =$

$y =$

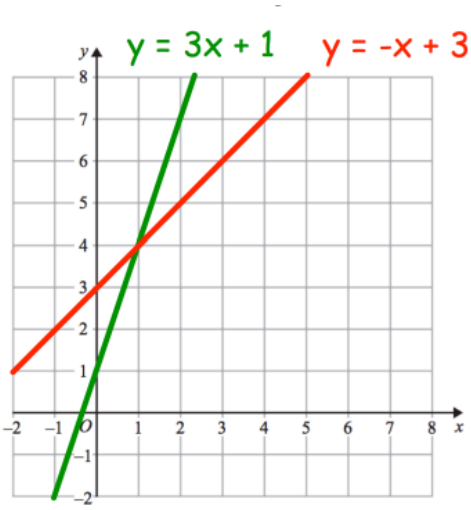
b)



$x =$

$y =$

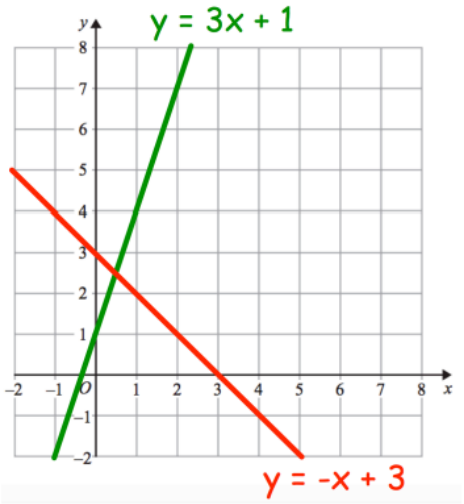
c)



$x =$

$y =$

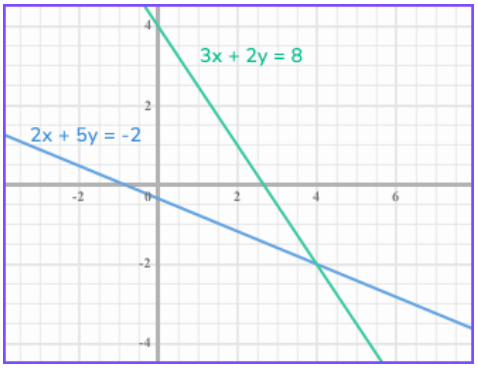
d)



$x =$

$y =$

e)

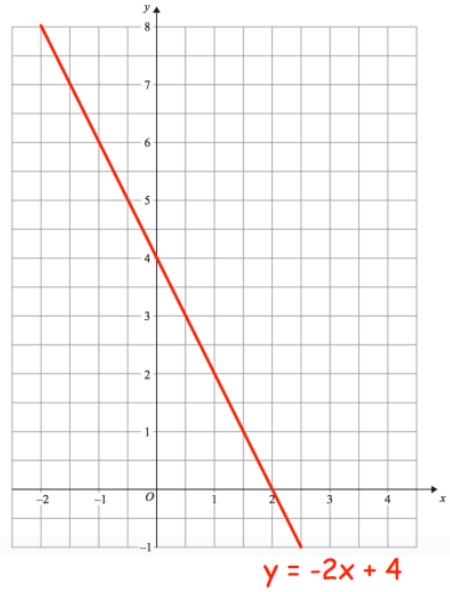


$x =$

$y =$

Example 5 - I do…

The straight line $y + 2x = 4$ has been drawn on the grid.



a) On the same grid, draw the graph of $y = x + 1$

x	-2	-1	0	1	2
y					

b) Use the graphs to solve the simultaneous equations

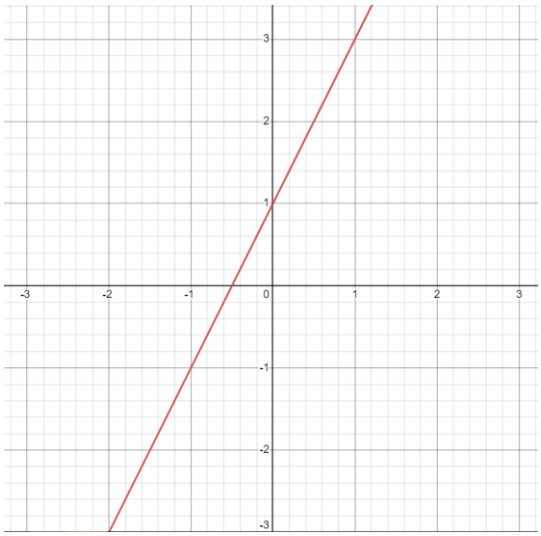
$y + 2x = 4$
$y = x + 1$

$x =$

$y =$

Example 6 – We do…

The straight line $y = 2x + 1$ has been drawn on the grid



a) On the same grid, draw the graph of $y = x$

x	-2	-1	0	1	2
y					

b) Use the graphs to solve the simultaneous equations

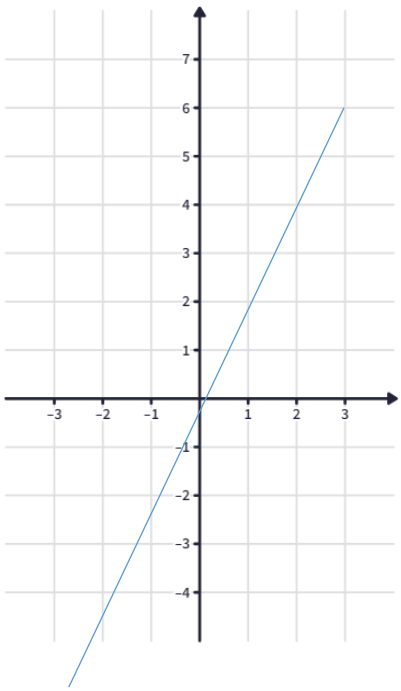
$y = 2x + 1$
$y = x$

$x =$

$y =$

Example 7 - You do…

The straight line $y = 2x$ has been drawn on the grid.



a) On the same grid, draw the graph of $y = 3x - 1$

x	-2	-1	0	1	2
y					

b) Use the graphs to solve the simultaneous equations

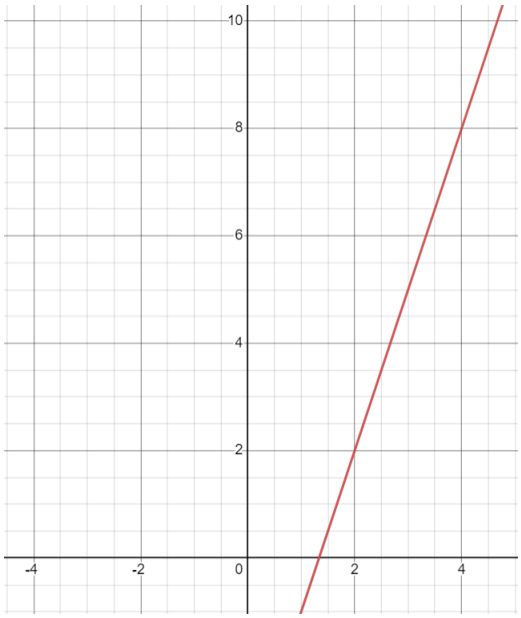
$y = 2x$
$y = 3x - 1$

$x =$

$y =$

Example 8 – You do…

The straight line $y = 3x - 4$ has been drawn on the grid



a) On the same grid, draw the graph of $y = x + 4$

x	0	1	2	3	4
y					

b) Use the graphs to solve the simultaneous equations

$y = 3x - 4$
$y = x + 4$

$x =$

$y =$

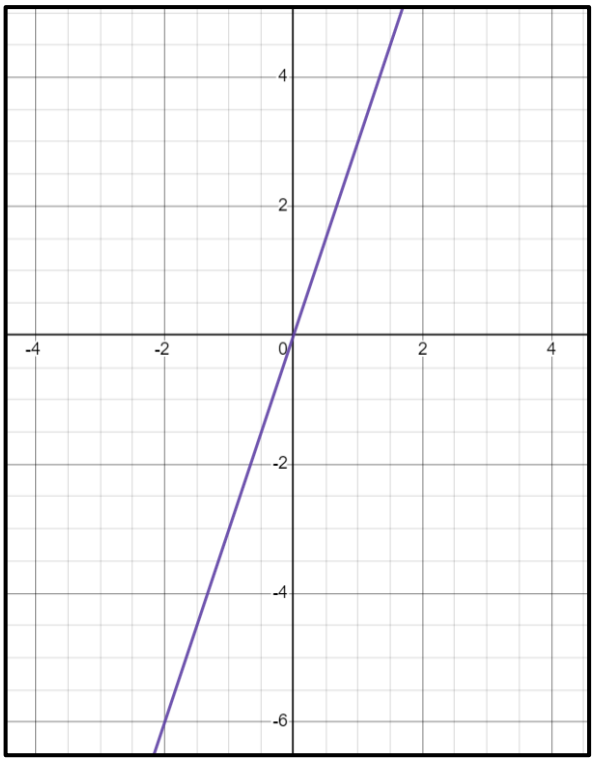
INDEPENDENT PRACTICE

Solve the simultaneous equations using the graphical method:

2a) The straight line $y = 3x$ has been drawn on the grid.

On the same grid draw the graph of
$y = 2x + 1$

x	-2	-1	0	1	2
y					



b) use the graph to find the solutions to the simultaneous equations $y = 3x$ and
$y = x - 2$

$x =$

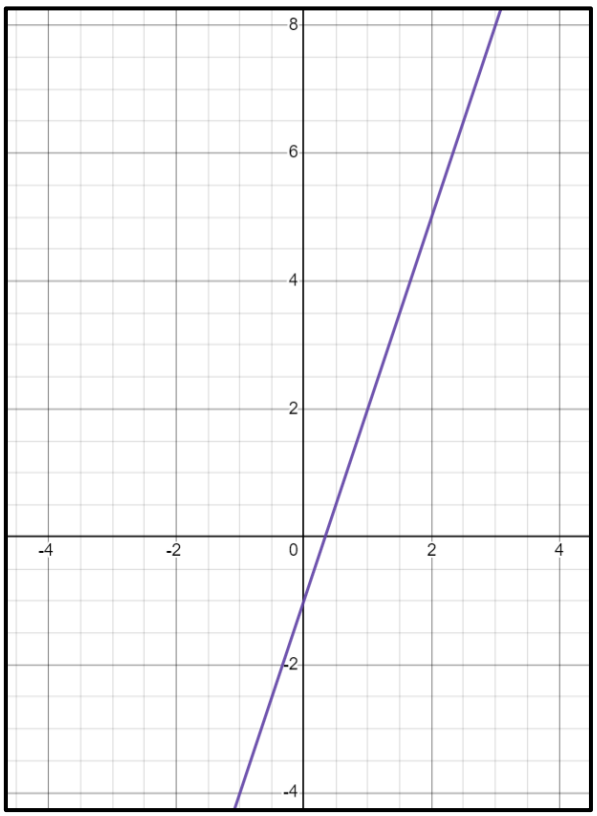
$y =$

Intersection points: ,

3a) The straight line $y = 3x - 1$ has been drawn on the grid.

On the same grid draw the graph of
$y = 2x + 1$

x	-2	-1	0	1	2
y					



b) use the graph to find the solutions to the simultaneous equations $y = 3x - 1$ and
$y = 2x + 1$

$x =$

$y =$

Intersection point: ,

Subtopic 3b – Recap: Simultaneous equations - Elimination

LEARNING OUTCOME: To recap solving simultaneous equations using the elimination method

Objectives:

1. To be able to eliminate variables by adding or subtracting equations.
2. To solve an equation with two variables by substituting a known variable.
3. Solve simultaneous equations using the elimination method

Keywords:
- Simultaneous equations
- Linear equations
- Equivalent equations
- Intersection
- Elimination

Elimination Method

Definition

A method to eliminate one of the variables in a pair of simultaneous equations to solve for both equations.

Characteristics

You must **equate** the variables for the elimination method to work.

Examples

$$\begin{array}{l} 5x + y = 11 \\ 4x + y = 9 \\ \hline x \quad\quad = 2 \end{array} -$$

$$\begin{array}{r} 5(2) + y = 11 \\ 10 + y = 11 \\ -10 \quad\quad -10 \\ y = 1 \end{array}$$

Non - Examples

$$\begin{array}{l} 5x + y = 11 \\ 4x + y = 9 \\ \hline 9x + 2y = 20 \end{array} +$$

$$\begin{array}{l} 5x + y = 11 \\ 4x - y = 9 \\ \hline x \quad\quad = 2 \end{array} -$$

Example 1 - I do…	*Example 2 – We do…*	*Example 3 – You do…*
Subtract equation 2 from equation 1. $3x + y = 10$ (1) $x + y = 5$ (2)	Subtract equation 2 from equation 1. $5x + 2y = 9$ (1) $3x + 2y = 5$ (2)	Subtract equation 2 from equation 1. $7x + 5y = 10$ (1) $3x + 5y = 5$ (2)

INDEPENDENT PRACTICE

1) subtract equation 2 from equation 1

1) $2x + y = 10$ (1)
$x + y = 5$ (2)

2) $8x + y = 10$ (1)
$5x + y = 5$ (2)

3) $3x + 3y = 10$ (1)
$x + 3y = 5$ (2)

4) $10x + 2y = 10$ (1)
$4x + 2y = 5$ (2)

5) $3x + 4y = 10$ (1)
$3x + y = 5$ (2)

6) $5x + 3y = 10$ (1)
$5x + y = 5$ (2)

7) $7x + 3y = 10$ (1)
$7x + y = 5$ (2)

8) $10x + 7y = 10$ (1)
$10x + 2y = 5$ (2)

Steps	*Example 7 - I do…*	*Example 8 – We do…*	*Example 9 – You do…*
1) Label the equations 1 and 2.	Solve the simultaneous equations $2x + y = 4$ (1) $x + y = 3$ (2)	Solve the simultaneous equations $3x + 2y = 9$ (1) $2x + 2y = 8$ (2)	Solve the simultaneous equations. $4x + 5y = 24$ (1) $3x + 5y = 23$ (2)
2) If the signs are the same for the variables that match subtract.			
3) Substitute the value of x or y into one of the equations to find the other values.			
4) Write the solutions.			

INDEPENDENT PRACTICE

3) Solve the simultaneous equations using elimination

a) $2x + y = 5$ (1)
$x + y = 4$ (2)

$x =$ ☐
$y =$ ☐

b) $2x + y = 4$ (1)
$x + y = 3$ (2)

$x =$ ☐
$y =$ ☐

c) $2x + y = 8$ (1)
$x + y = 6$ (2)

$x =$ ☐
$y =$ ☐

d) $3x + 2y = 29$ (1)
$2x + 2y = 24$ (2)

$x =$ ☐
$y =$ ☐

e) $4x + 3y = 33$ (1)
$3x + 3y = 30$ (2)

$x =$ ☐
$y =$ ☐

f) $5x + 4y = 60$ (1)
$4x + 4y = 56$ (2)

$x =$ ☐
$y =$ ☐

Steps	*Example 10 - I do…*	*Example 11 – We do…*	*Example 12 – You do…*
1) Label the equations 1 and 2.	Solve the simultaneous equations $4x + y = 6$ (1) $2x + y = 4$ (2)	Solve the simultaneous equations $6x + 5y = 27$ (1) $2x + 5y = 19$ (2)	Solve the simultaneous equations. $5x + 2y = 11$ (1) $2x + 2y = 8$ (2)
2) If the signs are the same for the variables that match subtract.			
3) Solve to find one variable.			
4) Substitute the value of x or y into one of the equations to find the other values.			
5) Write the solutions.			

INDEPENDENT PRACTICE

4) Solve the simultaneous equations using elimination

a) $3x + y = 6$ (1)
$x + y = 4$ (2)

$x =$ ☐

$y =$ ☐

b) $5x + y = 13$ (1)
$3x + y = 9$ (2)

$x =$ ☐

$y =$ ☐

c) $4x + y = 19$ (1)
$x + y = 7$ (2)

$x =$ ☐

$y =$ ☐

Steps	*Example 13 - I do…*	*Example 14 – We do…*	*Example 15 – You do…*
1) Label the equations 1 and 2.	Solve the simultaneous equations $4x - y = 6$ (1) $2x - y = 4$ (2)	Solve the simultaneous equations $5x - 2y = 11$ (1) $2x - 2y = 8$ (2)	Solve the simultaneous equations. $6x - 5y = 27$ (1) $2x - 5y = 19$ (2)
2) If the signs are the same for the variables that match subtract.			
3) Solve to find one variable.			
4) Substitute the value of x or y into one of the equations to find the other variable.			
5) Write the solutions.			

INDEPENDENT PRACTICE

5) Solve the simultaneous equations using elimination

a) $3x - y = 6$ (1)
$x - y = 4$ (2)

$x =$ ☐

$y =$ ☐

b) $5x - y = 13$ (1)
$3x - y = 9$ (2)

$x =$ ☐

$y =$ ☐

c) $4x - y = 19$ (1)
$x - y = 7$ (2)

$x =$ ☐

$y =$ ☐

Example 16 - I do…	Example 17 – We do…	Example 18 – You do…
Add the equations together to eliminate the y's $2x + y = 7$ (1) $x - y = 2$ (2) + _____	Add the equations together to eliminate the y's $3x + 2y = 16$ (1) $2x - 2y = 4$ (2) + _____	Add the equations together to eliminate the y's $4x - y = 31$ (1) $3x + y = 18$ (2) + _____

INDEPENDENT PRACTICE

6) Add the equations together to eliminate the y's

a) $3x + y = 4$ (1)
$x - y = 0$ (2)
_____

b) $8x - y = 8$ (1)
$5x + y = 5$ (2)
_____

c) $3x - 3y = 3$ (1)
$x + 3y = 5$ (2)
_____

d) $10x - 2y = 8$ (1)
$4x + 2y = 6$ (2)
_____

e) $3x - y = 4$ (1)
$2x + y = 8$ (2)
_____

f) $4x + y = 14$ (1)
$5x - y = 13$ (2)
_____

g) $8x + 2y = 28$ (1)
$7x - 2y = 17$ (2)
_____

h) $8x + 7y = 23$ (1)
$10x - 7y = 13$ (2)
_____

i) $3x - 4y = -5$ (1)
$3x + 4y = 11$ (2)
_____

j) $5x - 8y = -11$ (1)
$5x + 8y = 21$ (2)
_____

k) $7x + 9y = -5$ (1)
$7x - 9y = -23$ (2)
_____

l) $10x + 10y = -50$ (1)
$10x - 10y = -70$ (2)
_____

d) $3x + 2y = 22$ (1)
$2x - 2y = 8$ (2)

$x =$ ☐

$y =$ ☐

e) $5x - 2y = 22$ (1)
$2x + 2y = 20$ (2)

$x =$ ☐

$y =$ ☐

f) $4x + 3y = 50$ (1)
$2x - 3y = -2$ (2)

$x =$ ☐

$y =$ ☐

g) $5x + 2y = 23$ (1)
$2x - 2y = -2$ (2)

$x =$ ☐

$y =$ ☐

h) $5x - 3y = 0$ (1)
$3x + 3y = 24$ (2)

$x =$ ☐

$y =$ ☐

i) $4x + 4y = 44$ (1)
$x - 4y = -24$ (2)

$x =$ ☐

$y =$ ☐

INDEPENDENT PRACTICE

8. To solve these equations should we add or subtract?
Circle either the + or -

1. $3x + 4y = 10$
$5x + 4y = 22$ + −

2. $3x + 4y = 10$
$5x + 4y = -22$ + −

3. $3x + 4y = 10$
$-5x + 4y = -22$ + −

4. $3x + 4y = 10$
$-5x - 4y = -22$ + −

5. $3x + 4y = -10$
$-5x - 4y = -22$ + −

6. $-3x + 4y = -10$
$-5x - 4y = -22$ + −

7. $-3x - 4y = -10$
$-5x - 4y = -22$ + −

8. $-3x - 4y = -10$
$-5x + 4y = -22$ + −

9. $5x - 2y = -10$
$5x + 4y = -22$ + −

10. $-5x - 2y = -10$
$5x + 4y = -22$ + −

11. $5x - 2y = -10$
$-5x + 4y = -22$ + −

12. $-5x - 2y = -10$
$-5x + 4y = -22$ + −

9. Solve the simultaneous equations

a) $2x + 3y = 18$
$6x + 3y = 30$

x = ☐ , y = ☐

b) $2x + y = 18$
$6x - y = 30$

x = ☐ , y = ☐

c) $2x + 3y = 18$
$6x - 3y = 30$

x = ☐ , y = ☐

(d) $3x - y = 3$
$5x + y = 13$

x = , y =

(e) $x - y = 9$
$3x + y = 11$

x = , y =

(f) $2x - y = 7$
$3x + y = 8$

x = , y =

(g) $2x - 2y = 5$
$x - 2y = 3$

x = , y =

(h) $3x - 4y = 15$
$x - 4y = 3$

x = , y =

(i) $3x - 4y = 20$
$x - 4y = 8$

x = , y =

LEARNING OUTCOME: To solve simultaneous equations using the elimination method

Objectives:

1. Finding the LCM between two numbers.
2. Solving simultaneous equations by adjusting one equation.
3. Solving simultaneous equations by adjusting both equations.

Keywords:
- Simultaneous equations
- Linear equations
- Equivalent equations
- Intersection
- Elimination

Simultaneous Equations

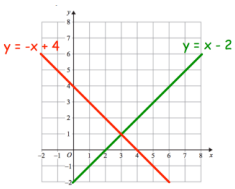
Definition

Two equations that form straight lines that **intersect at one point.**

Characteristics

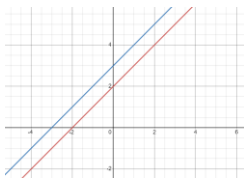
Linear simultaneous equations are two straight line equations that have an intersection point (this means they cross on the graph).

Examples



These are simultaneous equations because they **intersect** at a point.

Non - Examples



These equations do not intersect, therefore they are not simultaneous.

SKILLS CHECK

1) Find the lowest common multiple of these numbers

a) 3 and 9

b) 4 and 12

c) 10 and 30

d) 7 and 28

e) 21 and 42

Steps	*Example 1 - I do…*	*Example 2 – We do…*	*Example 3 – You do…*
1) Find the LCM of the variables.	Adjust one of the equations to make the $y's$ the same. $2x + y = 7$ (1) $x - 3y = 10$ (2)	Adjust one of the equations to make the $y's$ the same $3x + 4y = 16$ (1) $2x + 2y = 4$ (2)	Adjust one of the equations to make the $y's$ the same $4x - 21y = 31$ (1) $3x + 7y = 18$ (2)
2) Multiply one of the equations to make the variables the same.			

Adjusting one equation

Steps	*Example 4 - I do…*	*Example 5 – We do…*	*Example 6 – You do…*
1) Label the equations 1 and 2.	Solve the simultaneous equations $2x + y = 17$ (1) $3x + 2y = 29$ (2)	Solve the simultaneous equations $3x + y = 19$ (1) $6x + 3y = 42$ (2)	Solve the simultaneous equations. $4x + y = 11$ (1) $x + 3y = 11$ (2)
2) Find the LCM and adjust one of the equations.			
3) If the signs are **the same, subtract. If different, add.**			
4) Solve to find one variable.			
5) Substitute the value of x or y into one of the equations to find the other values.			
6) Write the solutions.			

INDEPENDENT PRACTICE

2) Solve the simultaneous equations using elimination

a) $2x + y = 5$ (1)
$3x + 2y = 9$ (2)

$x =$ ☐ , $y =$ ☐

b) $2x + 2y = 12$ (1)
$3x + 4y = 22$ (2)

$x =$ ☐ , $y =$ ☐

c) $2x + 3y = 28$ (1)
$3x + 6y = 51$ (2)

$x =$ ☐ , $y =$ ☐

Steps	*Example 7 - I do…*	*Example 8 – We do…*	*Example 9 – You do…*
1) Label the equations 1 and 2.	Solve the simultaneous equations $2x + y = 5$ (1) $3x - 2y = -3$ (2)	Solve the simultaneous equations $3x - y = 11$ (1) $6x + 3y = 27$ (2)	Solve the simultaneous equations. $4x + y = 22$ (1) $x - 3y = -1$ (2)
2) Find the LCM and adjust one of the equations.			
3) If the signs are **the same, subtract. If different, add.**			
4) Solve to find one variable.			
5) Substitute the value of x or y into one of the equations to find the other values.			
6) Write the solutions.			

INDEPENDENT PRACTICE

3) Solve the simultaneous equations using elimination

a) $2x - 2y = 6$ (1)
$3x + 4y = 16$ (2)

x = ☐ , y = ☐

b) $2x + 3y = 16$ (1)
$3x - 6y = 3$ (2)

x = ☐ , y = ☐

c) $2x - y = -1$ (1)
$3x + 2y = 9$ (2)

x = ☐ , y = ☐

MIXED PRACTICE

4) Solve the simultaneous equations using elimination

a) $2x + 8y = 43$ (1)
$x + 3y = 18$ (2)

$x =$ ☐ , $y =$ ☐

b) $5x - 3y = 33$ (1)
$3x - 9y = 63$ (2)

$x =$ ☐ , $y =$ ☐

c) $2x + 4y = -2$ (1)
$4x + 2y = -10$ (2)

$x =$ ☐ , $y =$ ☐

d) $8x + 4y = -28$ (1)
$3x - 12y = 30$ (2)

$x =$ ☐ , $y =$ ☐

e) $15x - 4y = 82$ (1)
$5x - 9y = 12$ (2)

$x =$ ☐ , $y =$ ☐

f) $12x + 3y = 9$ (1)
$2x + 11y = -9$ (2)

$x =$ ☐ , $y =$ ☐

Adjusting both equations

Steps	*Example 10 - I do…*	*Example 11 – We do…*	*Example 12 – You do…*
1) Label the equations 1 and 2.	Solve the simultaneous equations $4x + 2y = 8$ (1) $3x + 3y = 9$ (2)	Solve the simultaneous equations $3x + 2y = 13$ (1) $6x - 5y = 8$ (2)	Solve the simultaneous equations. $4x + 3y = 29$ (1) $x + 5y = 20$ (2)
2) Find the LCM and adjust **both** of the equations.			
3) If the signs are **the same, subtract. If different, add.**			
4) Solve to find one variable.			
5) Substitute the value of x or y into one of the equations to find the other values.			
6) Write the solutions.			

6) Solve the simultaneous equations by adjusting both equations

a) $2x + 2y = 14$ (1)
$5x - 3y = 19$ (2)

b) $2x + 3y = 1$ (1)
$7x + 2y = -22$ (2)

c) $5x + 3y = -4$ (1)
$2x + 4y = 4$ (2)

g) $3x + 8y = 31$ (1)
$5x + 3y = 31$ (2)

x = , y =

h) $7x - 15y = 2.5$ (1)
$3x - 2y = 5.5$ (2)

x = , y =

i) $3x + 2y = 53$ (1)
$2x + 5y = 72$ (2)

x = , y =

j) $5x - 3y = 18$ (1)
$2x + 4y = 54$ (2)

x = , y =

k) $2x + 8y = 8$ (1)
$9x + 3y = -63$ (2)

x = , y =

l) $2x - 4y = 4$ (1)
$5x - 3y = 24$ (2)

x = , y =

3) Solve the simultaneous equations by rearranging and then using elimination.

a) $x = 10 - y$ (1)
$2x + y = 17$ (2)

$x =$ ☐ , $y =$ ☐

b) $x - 4 = y$ (1)
$x + 3y = 12$ (2)

$x =$ ☐ , $y =$ ☐

c) $2x + 6y = 4$ (1)
$x = 12 + 2y$ (2)

$x =$ ☐ , $y =$ ☐

4) Two cats and a dog cost £91.
Three cats and two dogs cost £159.
How much does a cat cost?

Hint: **form equations**

Subtopic 3c – Simultaneous equations – Worded Problems

LEARNING OUTCOME: To solve simultaneous equations in worded problems

Objectives:

1. To recognise simultaneous equations in worded problems.
2. To be able to write worded problems as two coherent simultaneous equations, and solve these.
3. To be able to interpret the answers to worded simultaneous equations problems.

Keywords:

- Simultaneous equations
- Linear equations
- Equivalent equations

Simultaneous equations

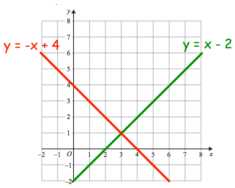
Definition

Two equations that form straight lines that **intersect at one point.**

Characteristics

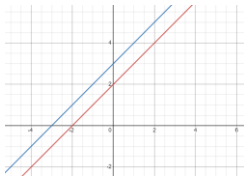
Linear simultaneous equations are two straight line equations that have an intersection point (this means they cross on the graph).

Examples



These are simultaneous equations because they **intersect** at a point.

Non - Examples



These equations do not intersect, therefore they are not simultaneous.

Key steps

1) Convert each sentence to an equation.
2) Solve the simultaneous equations.
3) Answer the question.

Example 1 - I do…

David buys 2 scones and 2 coffees in a shop and the cost is £18.
Eliza buys 3 scones and 2 coffees in the same shop and they cost £22.
Find the cost of each scone and each coffee.

Example 2 – We do…

The cost of buying a coffee and a tea in a café is £4.
The cost of buying a coffee and three teas in a café is £7.
Work out the cost of buying 2 coffees and the cost of buying 3 teas.

Example 3 – You do…

Five adult tickets and three child tickets for a movie cost £58.
Two adult tickets and eight child tickets for a movie cost £47.
Find the cost of each type of ticket.

INDEPENDENT PRACTICE 1. Solve the following

a) Sweets are sold in small packs and in big packs.
There is a total of 175 sweets in 4 small packs and 3 big packs.
There is a total of 154 sweets in 5 small packs and 2 big packs.
Work out the number of sweets in each small pack and in each big pack.

b) A plumber charges a price for each hour, £h, and a fixed charge, £c.
A 5 hour job costs £155 in total.
A 8 hour job costs £230 in total.
How much would a job that lasts 2 hours cost?

c) The sum of Rosemary's age and Hannah's age is 102 years.
The difference between Rosemary's age and Hannah's age is 52 years.
Rosemary is older than Hannah.
Find the age of each woman by using simultaneous equations.

3. Solve the following

a) Four sacks of potatoes and two sacks of carrots weigh 188 lb.
Five sacks of potatoes and one sack of carrots weigh 202 lb.
Baz buys seven sacks of potatoes and eight sacks of carrots.
Will he be able to carry them in his trailer, which can safely carry 450 lb?

b) In this sequence, the next term is found by multiplying the previous term by a and then adding b, where a and b are positive whole numbers.

3 14 47 ... ...

An equation that links the first and second terms is $3a + b = 14$.

a) Set up another equation in a and b.

b) Solve the equations to work out a and b.

c) Work out the next two terms in the sequence.

c) Five bags of bark and four trays of pansies cost £24.50.
One bag of bark and five trays of pansies cost £12.25.
Camilla wants six bags of bark and eight trays of pansies.
She has £30.
Will she have enough money?

4. Solve the following

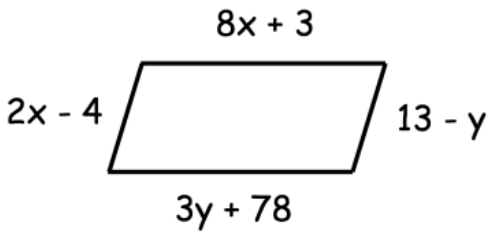
a) Albie is training for a marathon.
He jogs either route A or route B.

During April, he jogs route A nine times and route B five times.
Route B is 8 miles longer than route A.
In total, he jogs 89 miles in April.

In May, he will start jogging route C.
Route C is 20% longer than route B.

Work out the length of route C.

b) Shown below is parallelogram.
Each side is measured in centimetres.



Work out the perimeter of the parallelogram.

c) Solve the simultaneous equations

$$6x + 2y = 13c$$
$$x + 2y = -2c$$

Where c is a constant
Give your answer in terms of c.

GCSE QUESTIONS

Q1.
Solve the simultaneous equations

$$3x + y = -4$$
$$3x - 4y = 6$$

x = ...

y = ...

(Total for question = 3 marks)

Q2.
Solve the simultaneous equations

$$5x + y = 21$$
$$x - 3y = 9$$

x = ...

y = ...

(Total for question = 3 marks)

Q3.
3 kg of potatoes and 4 kg of carrots have a total cost of 440p.
4 kg of potatoes and 3 kg of carrots have a total cost of 470p.

Work out the total cost of 1 kg of potatoes and 1 kg of carrots.

..... p

(Total for question = 4 marks)

Q4.

3 teas and 2 coffees have a total cost of £7.80
5 teas and 4 coffees have a total cost of £14.20

Work out the cost of one tea and the cost of one coffee.

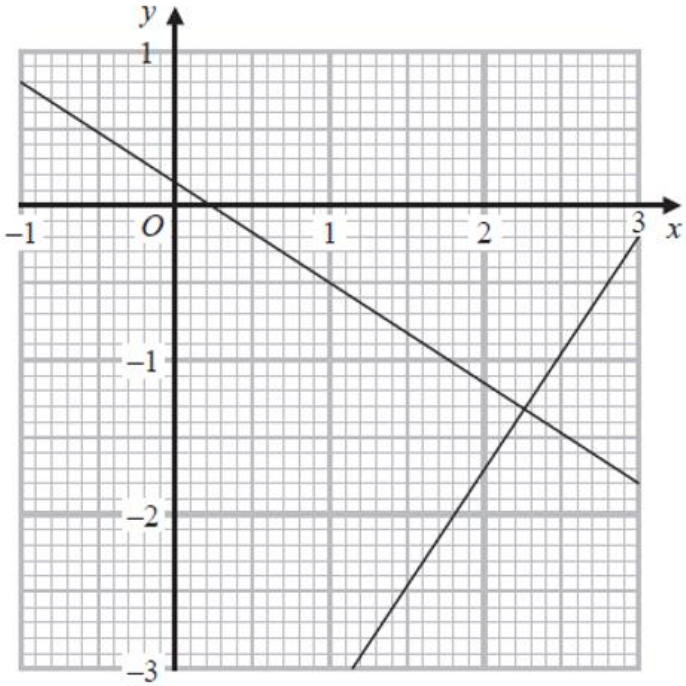
tea £ .....

coffee £ .....

(Total for question = 4 marks)

Q5.

The graphs with equations $3y + 2x = \frac{1}{2}$ and $2y - 3x = -\frac{113}{12}$ have been drawn on the grid below.



Using the graphs, find estimates of the solutions of the simultaneous equations

$$3y + 2x = \frac{1}{2}$$

$$2y - 3x = -\frac{113}{12}$$

x = ...

y = ...

(Total for question = 2 marks)

Q6.

Solve the simultaneous equations

$$5x + 2y = 11$$
$$4x + 3y = 6$$

x = ...

y = ...

(Total for question = 4 marks)

Q7.

Use algebra to solve the simultaneous equations

$$2x + 6y = 5$$

$$3x - 4y = -12$$

x= .....

y= .....

(Total for question = 4 marks)

Q8.

Use algebra to solve the simultaneous equations

$$4x - 5y = 20$$

$$6x + 7y = -57$$

You must show all your working.

x = ...

y = ...

(Total for question = 4 marks)

Q9.

Solve the simultaneous equations

$$5x - 2y = 23$$

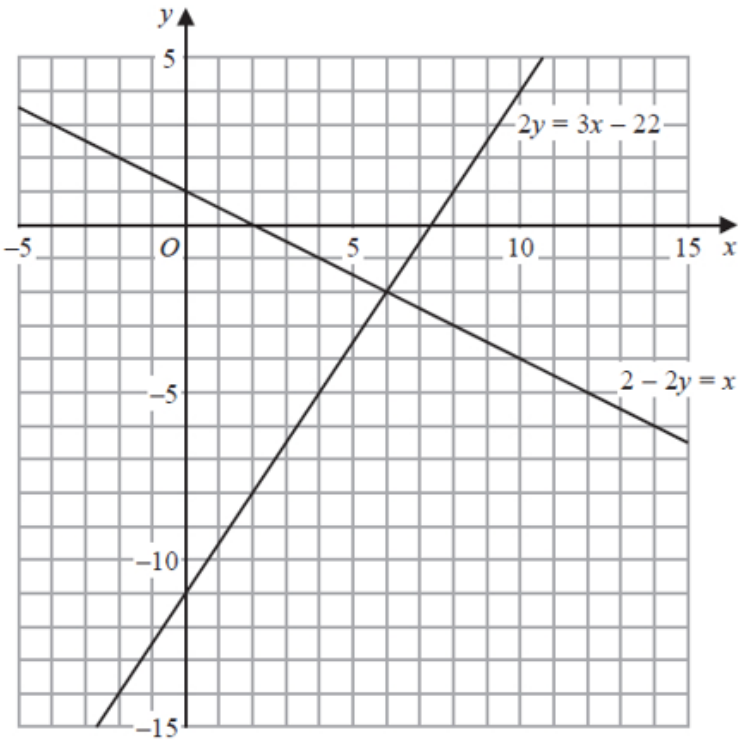
$$2x - 3y = 18$$

x = ...

y = ...

(Total for question = 4 marks)

Q10.



Use these graphs to solve the simultaneous equations

$$2 - 2y = x$$
$$2y = 3x - 22$$

x = ...

y = ...

(Total for question = 1 mark)

Subtopic 3d – Solving Simultaneous Equations by Substitution

LEARNING OUTCOME: To solve simultaneous equations using the substitution method

Objectives:

1. Solving simultaneous equations by substitution.
2. Solving simultaneous equations by adjusting and then substituting.

Keywords:
- Simultaneous equations
- Linear equations
- Equivalent equations
- Intersection
- Elimination

Substitution method

Definition

One equation is substituted into the other. This equation will often need to be expanded before it can be solved.

Characteristics

One equation may have x or y as it's subject.
e.g. $4x - y = 17$ and $y = x - 2$

Examples

Solve simultaneously

$$y = 2x + 1$$
$$5x + 3y = 14$$

Substitute $2x + 1$ for y into second equation

$$5x + 3(2x + 1) = 14$$

Expand and solve for x, before finding y.

Non - Examples

For these equations you would use the elimination method to solve simultaneously.

$$4x + y = 17$$
$$2x + y = 9$$

Steps	*Example 1 - I do…*	*Example 2 – We do…*	*Example 93– You do…*
1) Label the equations 1 and 2.	Solve the simultaneous equations $y = 2x - 3$ $5x - 3y = 11$	Solve the simultaneous equations $5x + 2y = 43$ $x = y - 4$	Solve the simultaneous equations. $4x + 3y = 88$ $y = 2x + 6$
2) Substitute isolated variable into equation 2.			
3) Expand			
4) Simplify			
5) Solve			
6) Substitute known variable to find the unknown variable.			

INDEPENDENT PRACTICE 1. Solve the simultaneous equations by substitution

a) $y = 2x$
$x + y = 6$

x = ☐ , y = ☐

b) $y = x + 2$
$2x + y = 11$

x = ☐ , y = ☐

c) $x = y + 3$
$x + 2y = 15$

x = ☐ , y = ☐

d) $x = 2y$
$7x - 2y = 60$

x = ☐ , y = ☐

$y = 3x$
$x + 2y = 14$

x = ☐ , y = ☐

f) $y = x + 3$
$y - 1 = 2x$

x = ☐ , y = ☐

Subtopic 3e – Quadratic simultaneous equations

LEARNING OUTCOME: To solve simultaneous equations using the substitution method

Objectives:

1. Finding the LCM between two numbers.
2. Solving simultaneous equations by adjusting one equation.
3. Solving simultaneous equations by adjusting both equations.

Keywords:
- Simultaneous equations
- Linear equations
- Equivalent equations
- Intersection
- Elimination

When solving simultaneous equations that involve a quadratic the elimination method will not be sufficient. Therefore, we need to use a substitution method. We will look to substitute a linear equation into the quadratic. As one equation is a quadratic you can receive two sets of answers to these equations. (i.e. 2 values for x and 2 values for y). It is important that you give these solutions as pairs.

Steps	*Example 1 - I do…*	*Example 2 – We do…*	*Example 3 – You do…*
1) Set each equation equal to each other.	Solve the simultaneous equations $y = x^2 + x - 2$ $y = 2x + 4$	Solve the simultaneous equations $y = x^2 + 7x - 2$ $y = x + 4$	Solve the simultaneous equations $y = x^2 - 3$ $y = 2x + 5$
2) Rearrange to = 0			
3) Solve by either: a) Factorising b) Quadratic Formula			
4) Substitute x values to find y			
5) Write solutions in pairs			

INDEPENDENT PRACTICE 2. Solve the quadratic simultaneous equations.

a) $y = x^2 + 5x - 2$
$y = x + 3$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

b) $y = x^2 - 9x - 3$
$y = x$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

c) $y = x^2 + 3x - 2$
$y = -x + 3$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

d) $y = x^2 - 3x - 2$
$y = x - 8$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

e) $y = x^2 - 5x - 2$
$y = -x + 3$

x = ☐ , y = ☐

x = ☐ , y = ☐

f) $y = x^2 - 3$
$y = x - 3$

x = ☐ , y = ☐

x = ☐ , y = ☐

g) $y = 2x^2 + x + 1$
$y = x^2 - 5x - 7$

x = ☐ , y = ☐

x = ☐ , y = ☐

h) $y = -x^2 + 5x + 2$
$y = 3x^2 - x - 2$

x = ☐ , y = ☐

x = ☐ , y = ☐

KNOWLEDGE CHECK Expand the following:

a) $(x+1)^2$	b) $(x-1)^2$	c) $(2x-3)^2$

Steps	*Example 4 - I do…*	*Example 5 – We do…*	*Example 6 – You do…*
1) Substitute linear equation into quadratic	Solve the simultaneous equations $x^2+y^2=9$ $y=x+3$	Solve the simultaneous equations $x^2+y^2=20$ $x=y-3$	Solve the simultaneous equations $x^2+y^2=17$ $x=y-5$
2) Expand and Simplify			
3) Rearrange to = 0			
3) Solve by either: a) Factorising b) Quadratic Formula			
4) Substitute x values to find y			
5) Write solutions in pairs			

INDEPENDENT PRACTICE 3. Solve the quadratic simultaneous equations.

a) $x^2 + y^2 = 25$
$y = x - 5$

$x = \square$, $y = \square$

$x = \square$, $y = \square$

b) $x^2 + y^2 = 4$
$x = y - 2$

$x = \square$, $y = \square$

$x = \square$, $y = \square$

c) $x^2 + y^2 = 20$
$y = x + 3$

$x = \square$, $y = \square$

$x = \square$, $y = \square$

d) $x^2 + y^2 = 25$
$y = 2x + 5$

$x = \square$, $y = \square$

$x = \square$, $y = \square$

KNOWLEDGE CHECK Expand the following:

a) $x(x-1)$	b) $x(2x-1)$	c) $4x(x-1)$

Steps	*Example 7 - I do…*	*Example 8 – We do…*	*Example 9 – You do…*
1) Substitute isolated equation into the quadratic	Solve the simultaneous equations $xy = 2$ $y = x + 1$	Solve the simultaneous equations $xy = 2$ $y = x - 1$	Solve the simultaneous equations $xy = 12$ $y = x + 1$
2) Expand and Simplify			
3) Rearrange to = 0			
3) Solve by either: a) Factorising b) Quadratic Formula			
4) Substitute x values to find y			
5) Write solutions in pairs			

INDEPENDENT PRACTICE 4. Solve the quadratic simultaneous equations.

a) $xy = 12$
$y = x - 1$

$x = \square$, $y = \square$

$x = \square$, $y = \square$

b) $xy = 12$
$y = x - 4$

$x = \square$, $y = \square$

$x = \square$, $y = \square$

c) $xy = 12$
$y = x + 4$

$x = \square$, $y = \square$

$x = \square$, $y = \square$

d) $xy = 12$
$y = 2x + 5$

$x = \square$, $y = \square$

$x = \square$, $y = \square$

MIXED PRACTICE

a) $xy = 24$
$y = 3x - 1$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

b) $xy = 12$
$2y = x + 5$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

c) $xy = 12$
$2y = x - 5$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

d) $4xy = 12$
$y = x - 2$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

e) $x^2 + y^2 = 25$
$y = 2x - 5$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

f) $2x^2 + y^2 = 17$
$y = x + 5$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

g) $2x^2 + y^2 = 10$
$y = 2x - 5$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

h) $x^2 + 2y^2 = 11$
$x = 5 - 2y$

$x =$ ☐ , $y =$ ☐

$x =$ ☐ , $y =$ ☐

5. Solve the quadratic simultaneous equations.

a) $y = x + 1$
$y = 1 + \frac{4}{x}$

$x = 2,\ y = 3$
$x = -2,\ y = -1$

$x =$ ☐ , $y =$ ☐
$x =$ ☐ , $y =$ ☐

b) $xy = 9$
$y = x$

1 $x = -3,\ y = -3$ and $x = 3,\ y = 3$

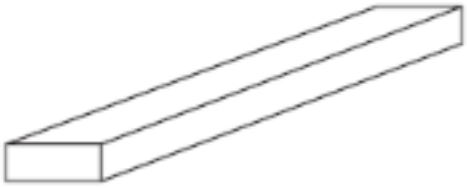
$x =$ ☐ , $y =$ ☐
$x =$ ☐ , $y =$ ☐

c) $x^2 + y^2 = 50$
$y = x$

2 $x = -5,\ y = -5$ and $x = 5,\ y = 5$

$x =$ ☐ , $y =$ ☐
$x =$ ☐ , $y =$ ☐

6**. A solid cuboid has a volume of 40 cm^3
The cuboid has a total surface area of 100 cm^2
One edge of the cuboid has length 2 cm.



Find the length of a diagonal of the cuboid.
Give your answer correct to 3 significant figures.

Q11.
Solve algebraically the simultaneous equations

$$2x^2 - y^2 = 17$$

$$x + 2y = 1$$

(Total for question = 5 marks)

Q12.
Solve the simultaneous equations

$$y^2 = 3x^2 + 4$$
$$y + 2x = 7$$

Give your solutions correct to 3 significant figures.

.....

(Total for question = 4 marks)

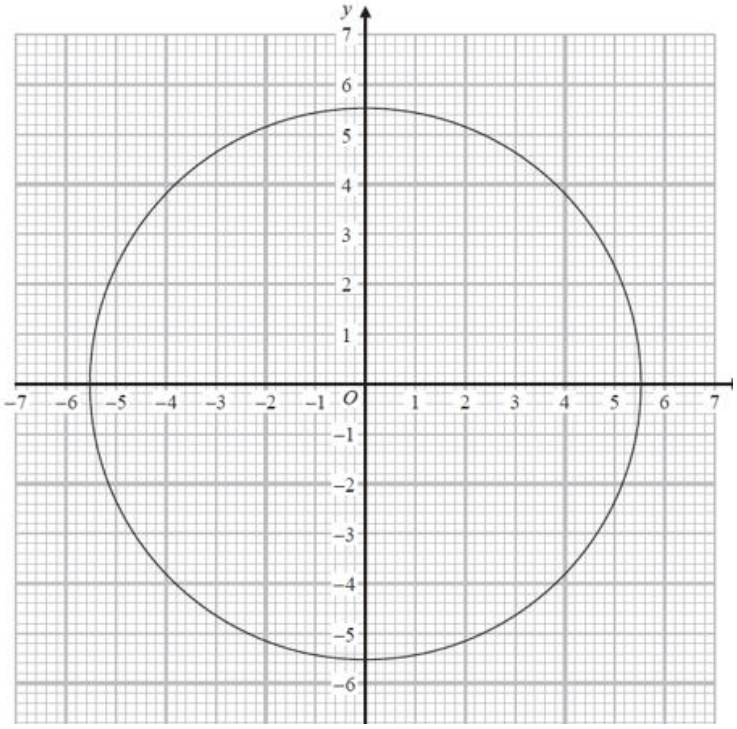
Q13.
Solve algebraically the simultaneous equations

$$x^2 + y^2 = 25$$
$$y - 3x = 13$$

.....

(Total for question = 5 marks)

Q14.
The diagram shows the graph of $x^2 + y^2 = 30.25$



Use the graph to find estimates for the solutions of the simultaneous equations

$$x^2 + y^2 = 30.25$$
$$y - 2x = 1$$

.....

(Total for question = 3 marks)

Q15.

The curve **C** has equation $y = x^2 + 3x - 3$

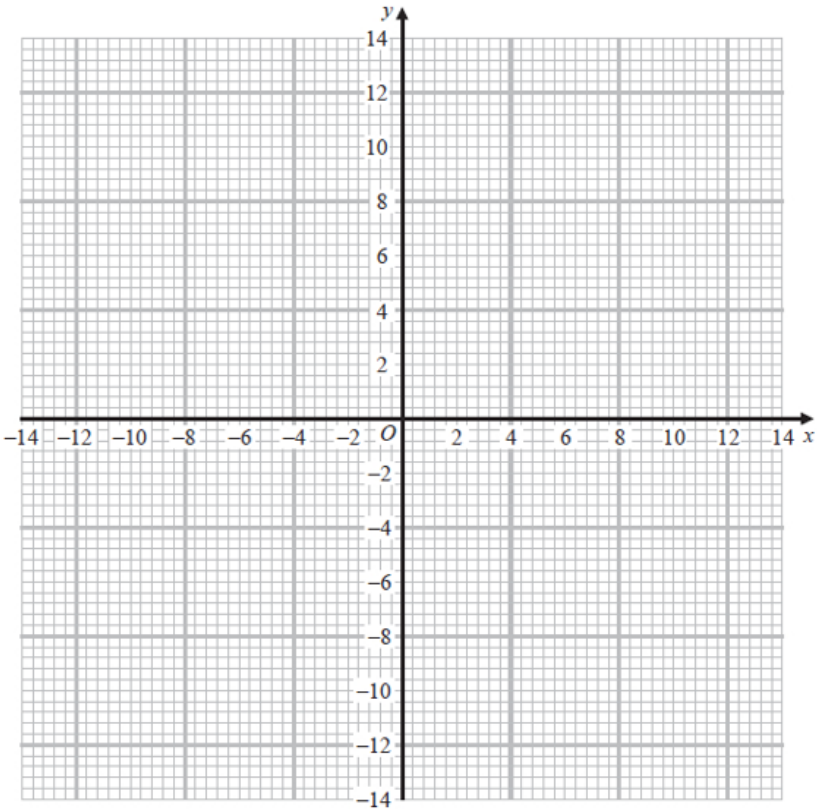
The line **L** has equation $y - 5x + 4 = 0$

Show, algebraically, that **C** and **L** have exactly one point in common.

(Total for question = 4 marks)

Q16.

(a) On the grid, draw the graph of $x^2 + y^2 = 169$



(2)

(b) Use your graph to find estimates for the solutions of the simultaneous equations

$$x^2 + y^2 = 169$$

$$2y = 3x$$

.....

(3)

(Total for question = 5 marks)